

# Project Coversheet

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|----------------------------------------------------------------|----------------------|
| Full Name                                                      | Purvi Rajes Tulcidas |
| Project Title<br>(Example –<br>Week1, Week2,<br>Week3, Week 4) | Week2                |

## Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

## Project Guidelines and Rules

### 1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

### 2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

### 3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

#### **4. Plagiarism Policy**

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

#### **5. Evaluation Criteria**

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

#### **6. Deadlines and Late Submissions**

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

#### **7. Additional Resources**

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

## Week 2: Project: Sales & Customer Behaviour Insights – Green Cart Ltd.

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### Introduction

This report is made for Green Cart Ltd., the analysis is made to gain insight into its recent sales, product and customer data to better understand revenue performance, customer behaviour and delivery outcomes. The dataset provided consists of three sources sales transactions, product information and customer profiles, which were cleaned, merged and analysed to create a unified view of business activity.

The purpose of this report is to identify key revenue trends across regions and product categories, examine how customer loyalty and discounts influence purchasing behaviour and assess delivery performance. By exploring these patterns, the analysis aims to highlight areas of strong performance as well as potential risks or opportunities for improvement. The findings are supported by summary tables and visualisations to present clear, actionable insights that can help the business make informed operational and strategic decisions.

### Data Cleaning Summary

Before analysis, the datasets were carefully cleaned to ensure accuracy and consistency. Text-based fields such as `delivery_status`, `payment_method`, `region`, `gender` and `loyalty_tier` was standardised by correcting inconsistent casing, removing extra whitespace and correcting misspellings. This ensured that categories such as “Delivered”, “Delayed” or “Gold” were treated consistently during aggregation.

Missing values were handled based on their business relevance. Discount values were filled with zero where no discount was applied, while records with missing critical identifiers such as `order_id`, `customer_id`, or `order_date` were removed to avoid

incorrect calculations. Duplicate sale records were identified using `order_id` and removed to prevent double-counting revenue.

During exploratory analysis, additional label inconsistencies, such as region and delivery status typos were identified post-merge and were corrected to ensure accurate summary tables and visualisations.

```
sales_df.isnull().sum()

0
order_id      0
customer_id   0
product_id    0
quantity      0
unit_price    0
order_date    0
delivery_status 0
payment_method 0
region        0
discount_applied 0
dtype: int64
```

Figure 1: checking for missing values after cleaning

```
#checking for duplicates on column that is for transaction
sales_df.duplicated(subset=['order_id']).sum()

np.int64(2)

#removing duplicates
sales_df = sales_df.drop_duplicates(subset=['order_id'])
```

Figure 2: checking and removing duplicates on `order_id`

## Feature Engineering Summary

Several new features were created to support deeper analysis and answer the business questions effectively. Revenue was calculated for each order by combining quantity, unit\_price, and any applied\_discount, allowing a more accurate measure of sales revenue. An order\_week variable was derived from the order\_date using ISO week numbering to enable weekly trend analysis across regions.

To compare delivery performance and pricing behaviour, unit prices were grouped into low, medium and high price\_band. A days\_to\_order feature was created to measure the time between a product's launch date and when it was purchased, providing insight into

product uptake speed. Customer email\_domain was extracted to support segmentation analysis and an is\_late flag was added to identify delayed deliveries. Together, these engineered features enabled meaningful aggregation, trend analysis and clear visualisation of key business metrics.

Key Findings & Trends

The analysis shows clear patterns in revenue performance across time, product categories and customers. Weekly revenue trends indicate that overall sales increased towards the middle of the analysis period, with several regions experiencing revenue peaks around the same weeks. This suggests revenue growth was driven by broader demand patterns rather than performance from a single region.

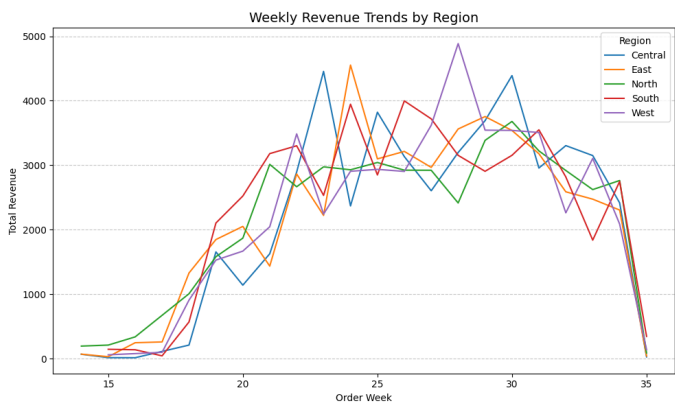


Figure 3: weekly revenue trends by region (line plot)

Product category analysis indicates that the “Cleaning” category consistently generated the highest revenue, followed by “Storage” and “Outdoors”. These categories contributed most strongly during high-revenue weeks, highlighting their importance in driving overall sales performance.

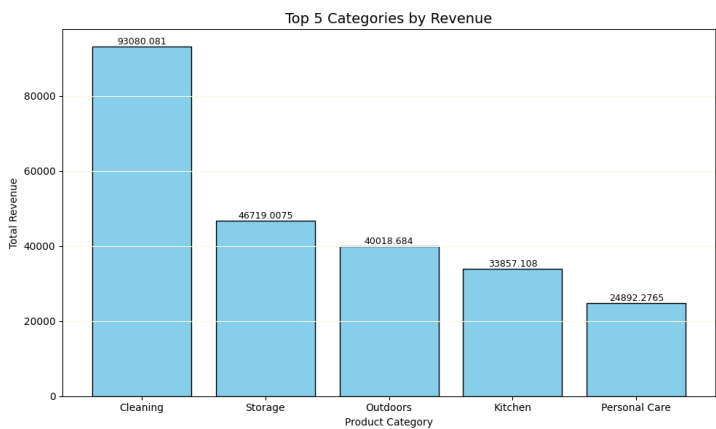


Figure 4: revenue by product category (bar chart)

Customer behaviour analysis further shows that “Gold” loyalty\_tier customers generate the greatest share of total revenue, indicating that higher-tier customers play a key role in orders leading to revenue growth and representing a valuable sector for targeted engagement and retention strategies.

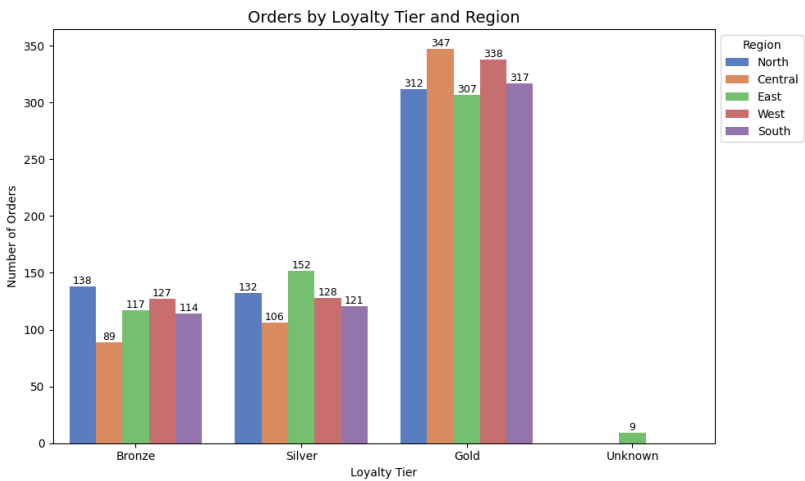


Figure 5: orders by revenue by loyalty tier and region (countplot)

Business Question Answers

1. Which product categories drive the most revenue, and in which regions?

“Cleaning” category generates the highest total revenue, followed by “Storage” and “Outdoors”. These categories consistently outperform others and are the primary drivers of sales. Revenue trends by region show all regions contribute to overall performance, with higher revenue peaks happening at similar times (around weeks 28-30) across regions rather than being driven by a single location, indicating strong-performing categories contribute broadly across regions rather than being concentrated in one area.

|   | category      | total_revenue | total_quantity | average_discount |
|---|---------------|---------------|----------------|------------------|
| 0 | Cleaning      | 93080.0810    | 3566.0         | 0.085577         |
| 4 | Storage       | 46719.0075    | 1725.0         | 0.081043         |
| 2 | Outdoors      | 40018.6840    | 1515.0         | 0.081881         |
| 1 | Kitchen       | 33857.1080    | 1224.0         | 0.075436         |
| 3 | Personal Care | 24892.2765    | 900.0          | 0.087043         |

Figure 6: product category performance (revenue, quantity, discount (summary table)

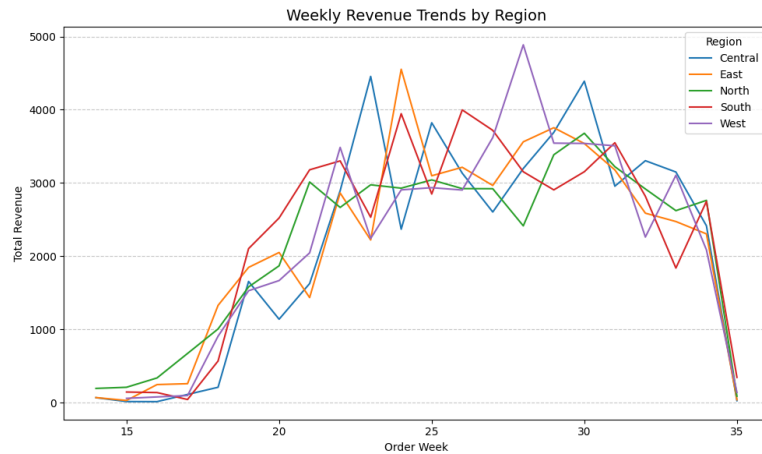


Figure 7: weekly revenue trends by region (line plot)

## 2. Do discounts lead to more items sold?

Orders with discounts show slightly higher quantities ordered compared to non-discount orders (low and medium discount ranges), median quantity remain 3 across all discount levels. However, it suggests that discounts may encourage some additional purchasing but are not the sole driver of higher order volumes. Other factors such as product type and customer loyalty, possibly influence purchasing behaviour.

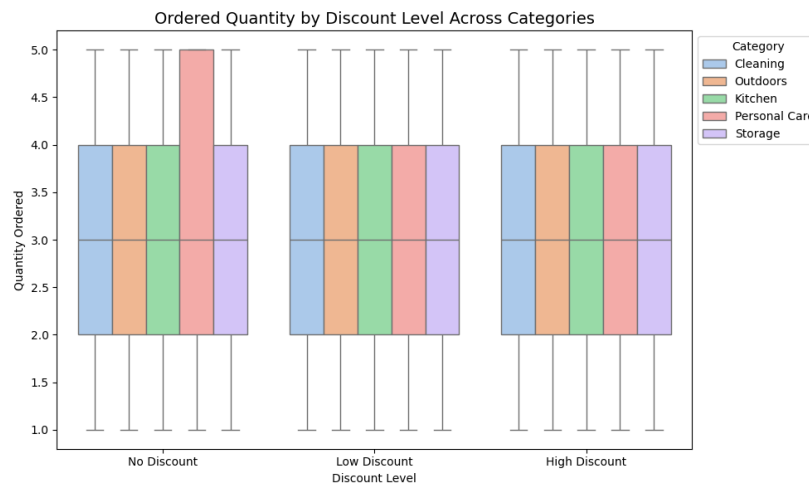


Figure 8: comparison of quantity across discount levels (boxplot)

## 3. Which loyalty tier generates the most value?

“Gold” loyalty customers generate highest total revenue, followed by “Silver” and “Bronze” customers, indicating that higher-tier loyalty members are more valuable to the business and contribute disproportionately to sales. Retaining and engaging these customers is vital to sustain revenue growth.

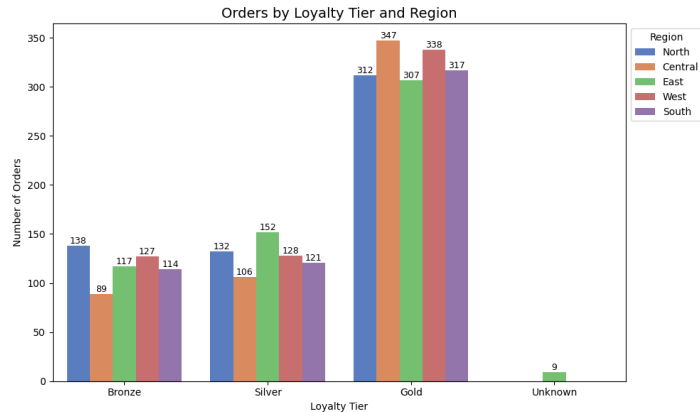


Figure 9: orders by revenue by loyalty tier (countplot)

#### 4. Are certain regions struggling with delivery delays?

Delivery delays occur across all regions, with no region showing disproportionately poor-performance. Delays are present across all price bands, suggesting that delivery challenges are not limited to a specific region or product price range, possible cause is due to operational factors rather than regional issues.

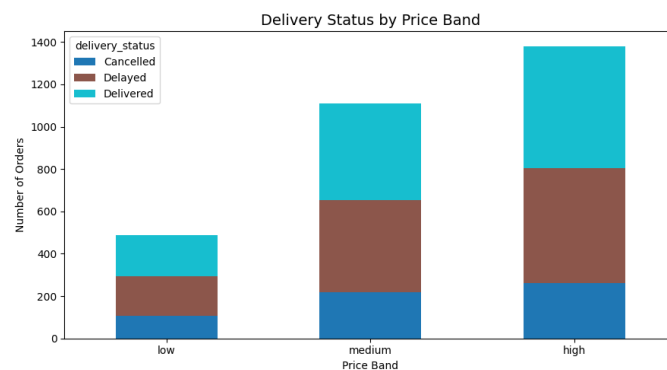


Figure 10: delivery status by price band (stacked bar chart)

#### 5. Do customer signup patterns influence purchasing activity?

Customer behaviour analysis shows customers who signed up earlier, in particular those who proceeded into higher loyalty tiers, tend to generate more revenue over time. While signup month does not directly determine purchasing activity, long-term engagement and loyalty status appear to have a stronger influence on customer value.

|    | loyalty_tier | signup_month | total_revenue | total_orders |
|----|--------------|--------------|---------------|--------------|
| 17 | Gold         | 2024-08      | 15645.4105    | 185          |
| 21 | Gold         | 2024-12      | 14059.3285    | 184          |
| 23 | Gold         | 2025-02      | 13220.9875    | 146          |
| 18 | Gold         | 2024-09      | 11978.9265    | 145          |
| 19 | Gold         | 2024-10      | 11582.0955    | 141          |

Figure 11: customer behaviour by loyalty tier and signup month (summary table)

## Recommendations

1. Company should prioritise high-performing product categories such as “Cleaning” and “Storage” by ensuring stock availability and targeted promotions. These are the categories that consistently generate high revenue and offer opportunity for short-term sales growth
2. Business should focus on retaining and rewarding “Gold” loyalty customers through personalised offers, early access to promotions or enhanced loyalty benefits, as this group contributes to largest share of revenue, improving engagement with these customers can support long-term profitability
3. Delivery performance should be reviewed across all regions and price bands to reduce delays. While no single region is disproportionately affected, consistent delays suggest operational improvements such as improved logistics coordination or clearer delivery timelines, these can enhance customer satisfaction and reduce potential churn

## Data Issues or Risks

One key data quality issue identified during the analysis was the presence of inconsistent and misspelled categorical values. Examples, include variations in delivery\_status such as “DELAYED”, “Delyd”, “Delrd”. And in loyalty\_tier “Gld”, “Brnze”, “Sllver”, these required manual standardisation before accurate aggregation could be performed. Moreover, some customer records contained missing values for loyalty tier and demographic fields.

To address these issues at the source, Green Cart Ltd., should implement stricter data validation rules, such as controlled dropdown selections for categorical fields and automated checks to flag invalid or missing entries. This would reduce manual cleaning effort and improve data reliability for future analysis.